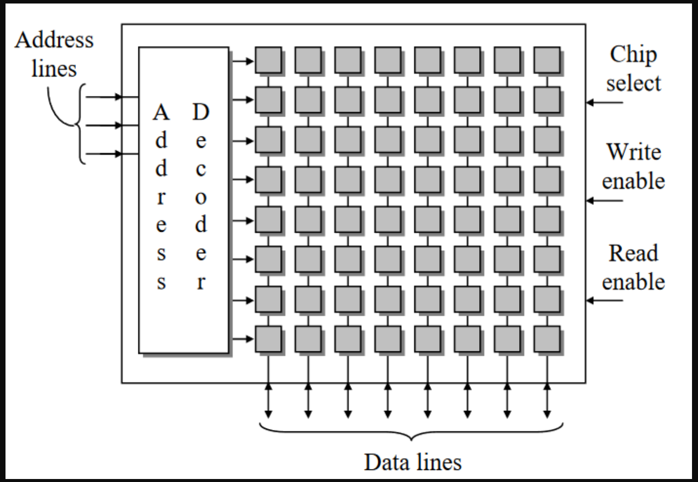
Lecture 7
Topics: Memory Organization



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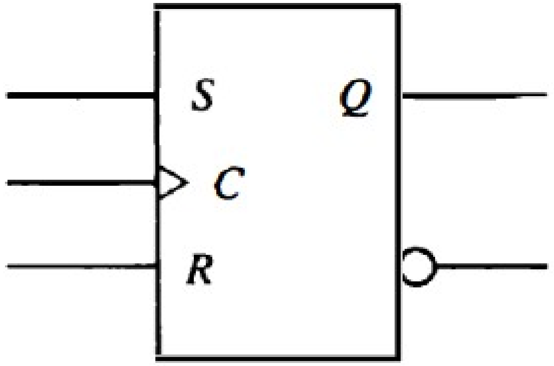
CIS-310: Computer Organization & Assembly Language

MW

Recap - flipflops

- SR flip-flop



(a) Graphic symbol

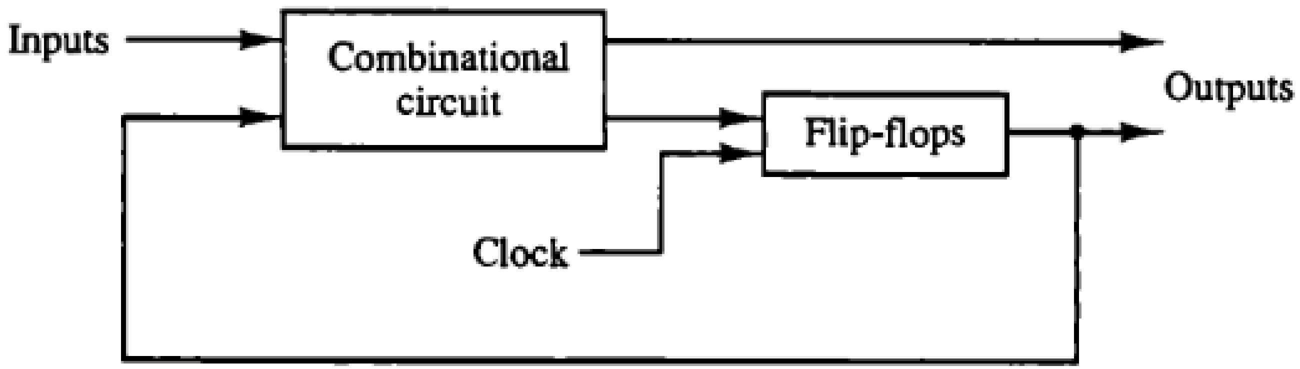
S	R	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Clear to 0
1	0	1	Set to 1
1	1	?	Indeterminate

(b) Characteristic table

- D flip-flop
- T flip-flop
- JK flip-flop
- ..

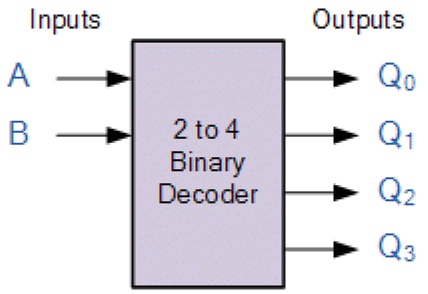
Recap – Sequential circuits

- Binary counter
- State machines
- …



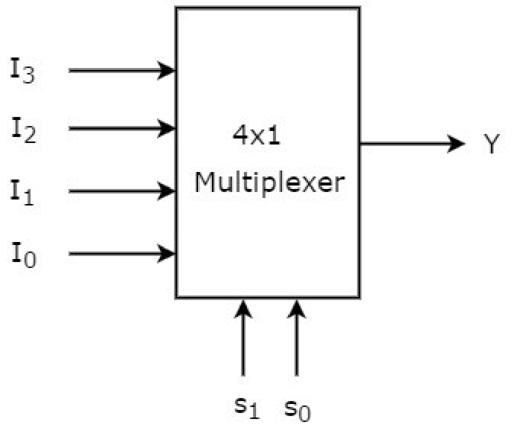
Recap – Combinational circuit

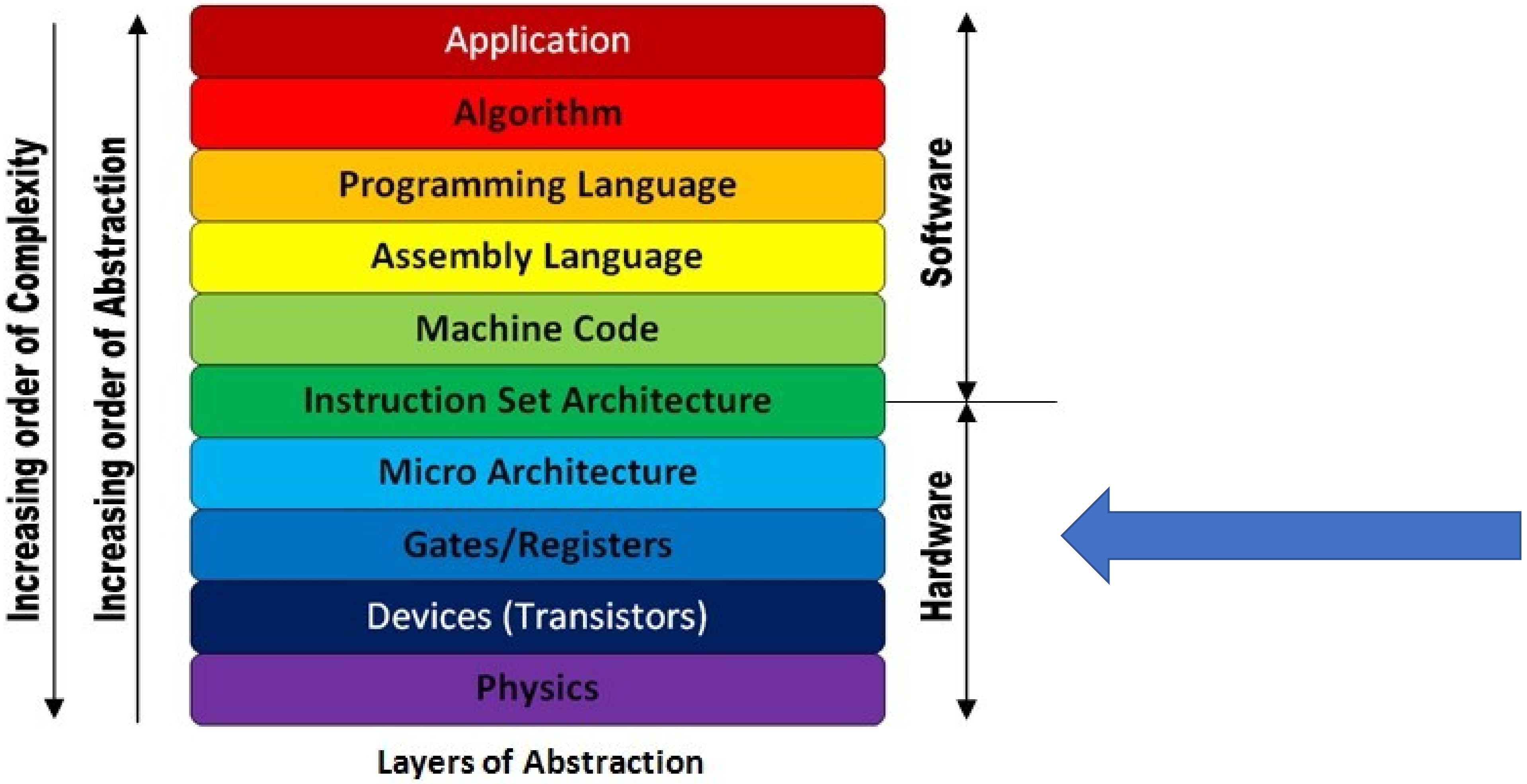
- Adder circuit
- Seven segment display
- Decoders
- Multiplexer



Truth Table

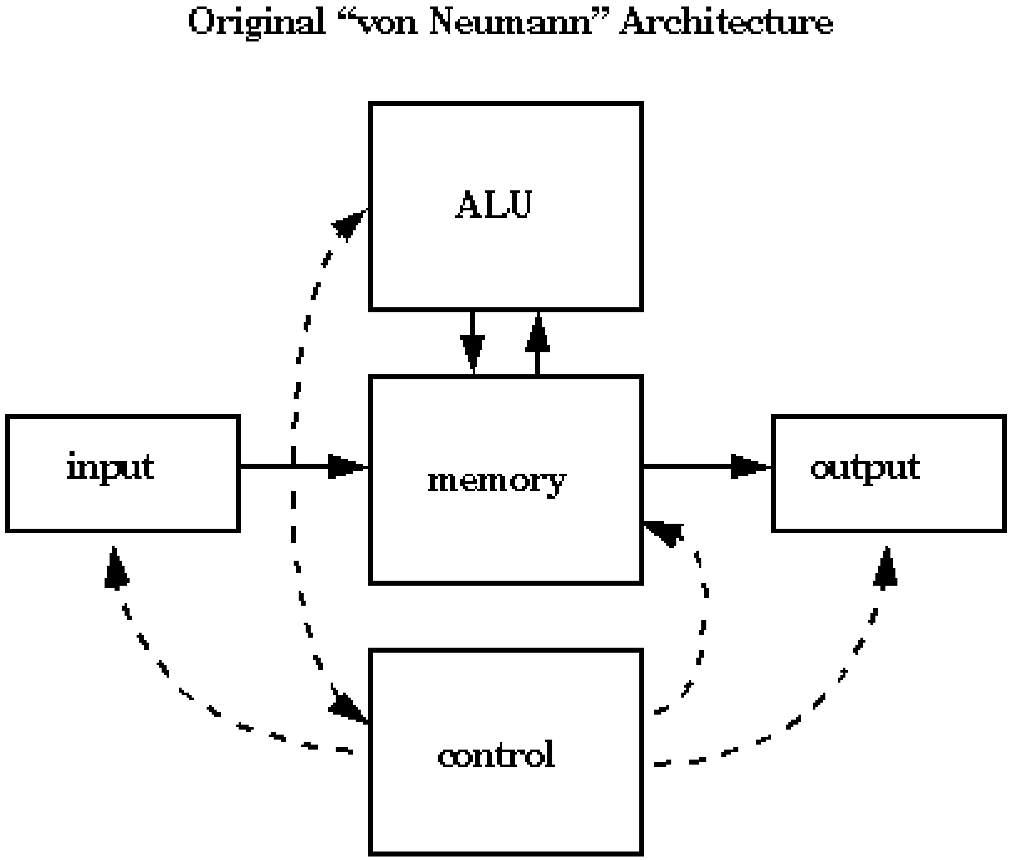
A	B	Q_0	Q_1	Q_2	Q_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



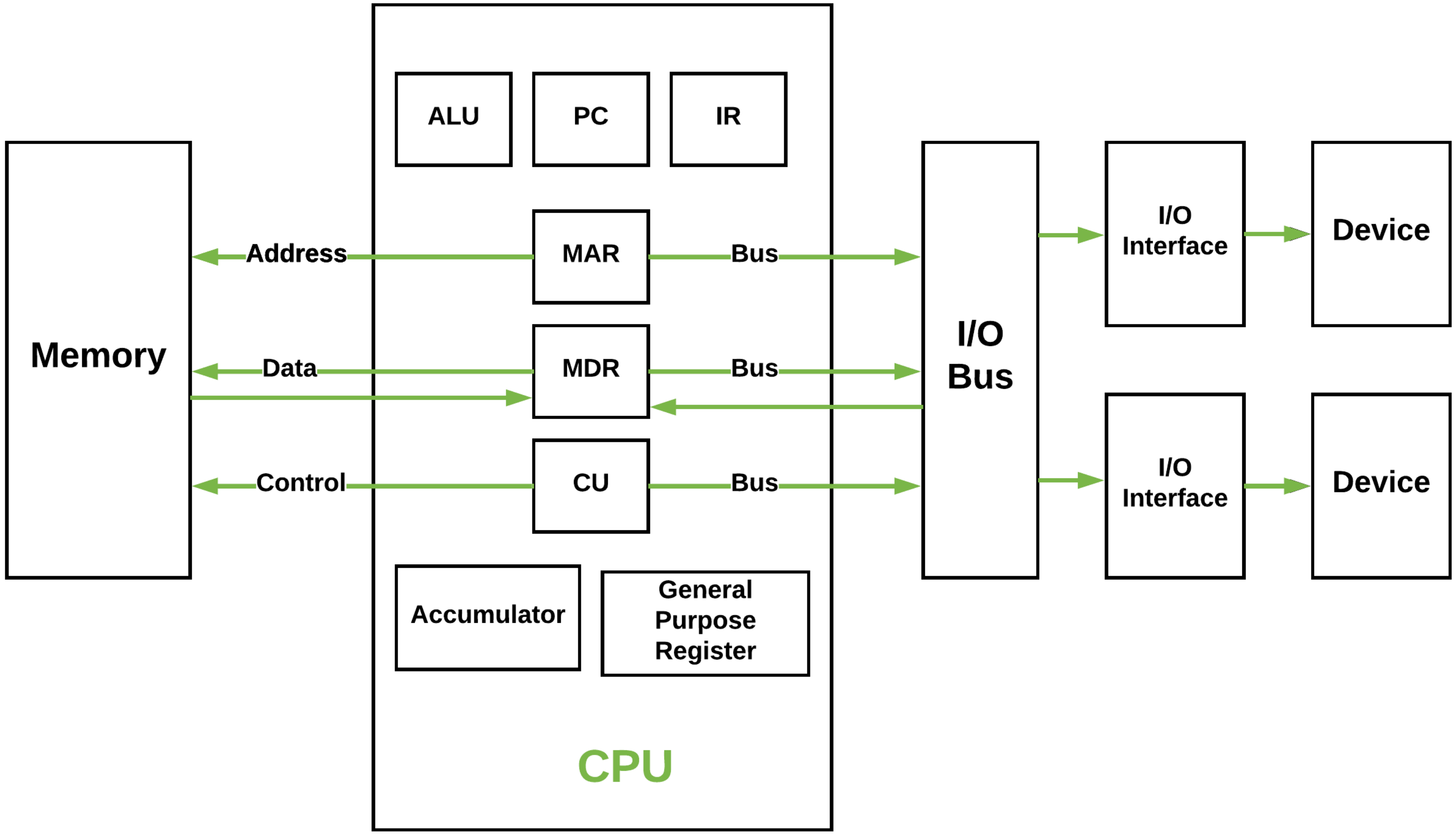


https://faculty.etsu.edu/tarnoff/ntes2150/Ch12_v02.pdf

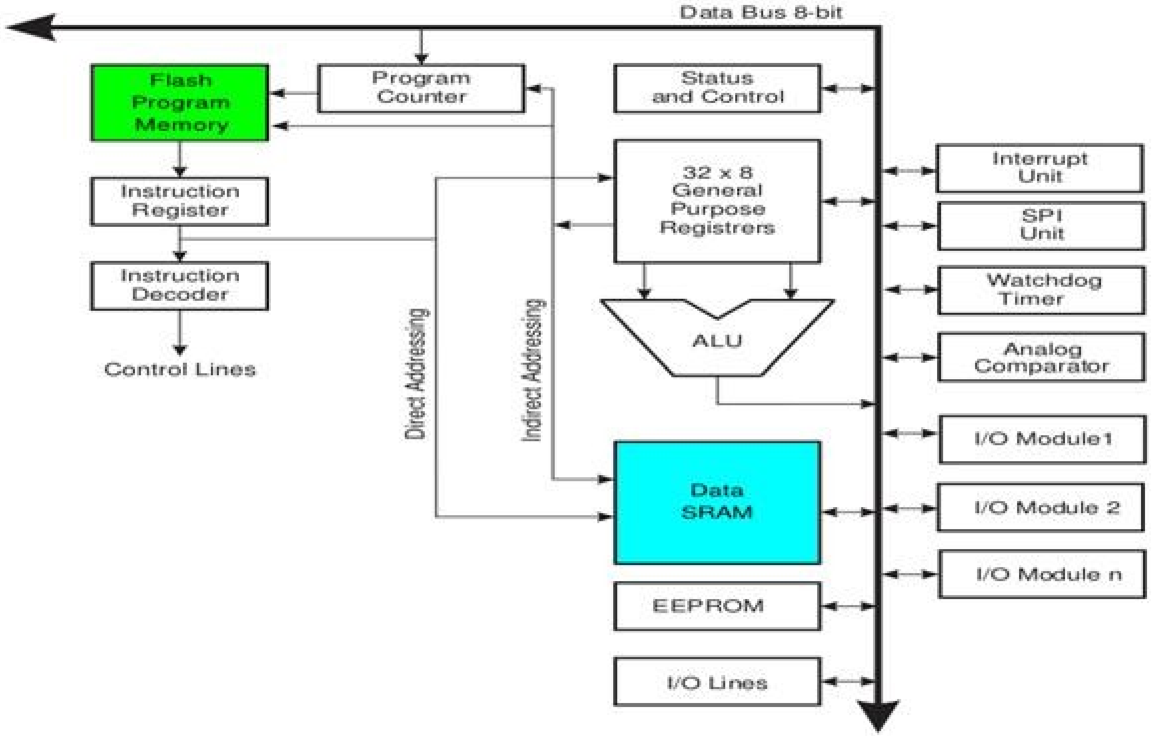
Computer architecture



Computer architecture – another view

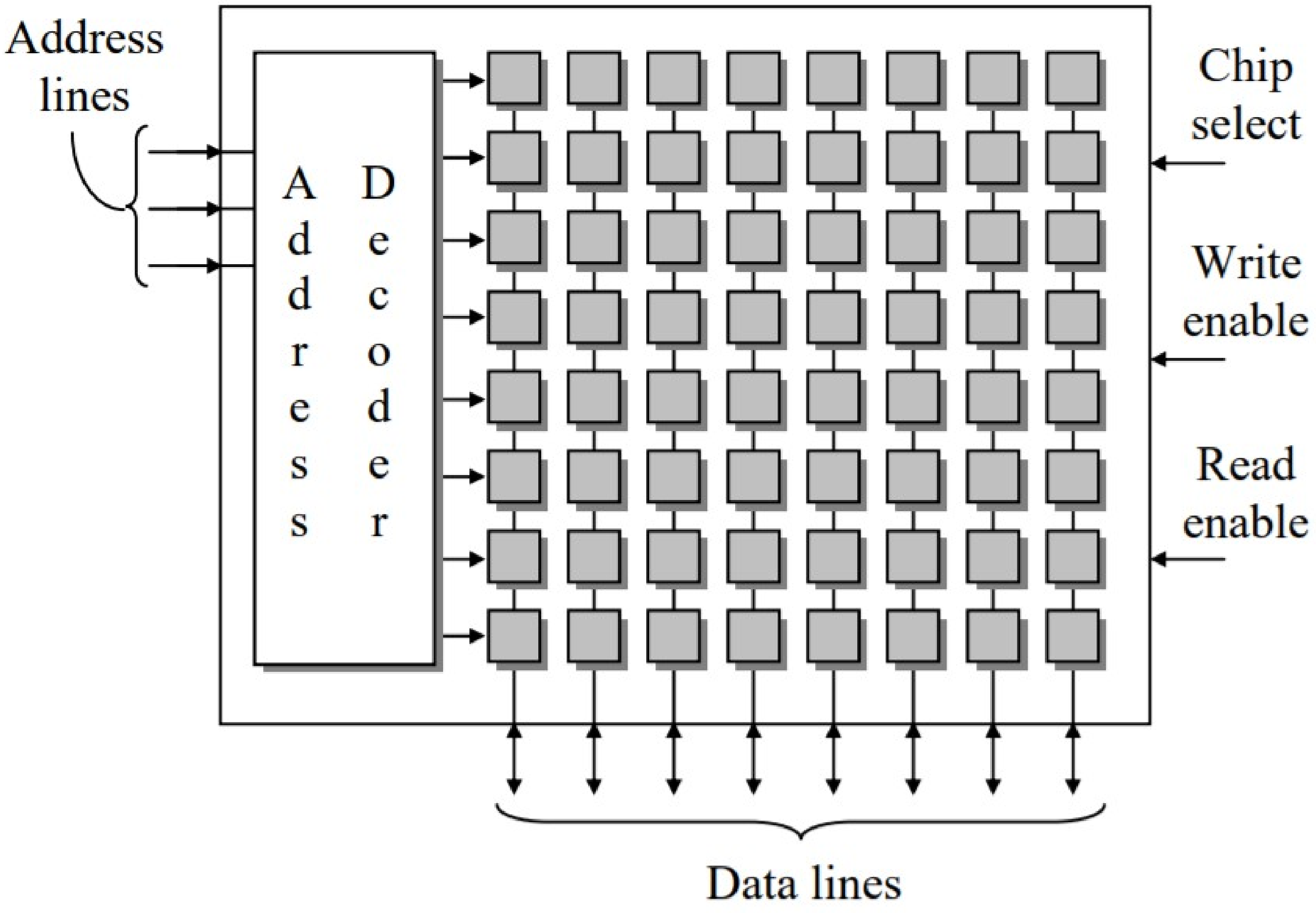


ATmega328 microcontroller Architecture (Arduino)



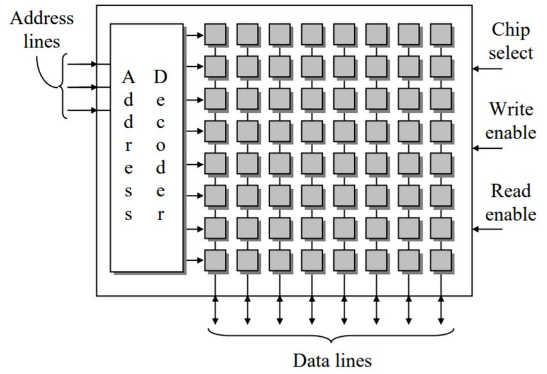
Harvard Architecture

Memory Organization



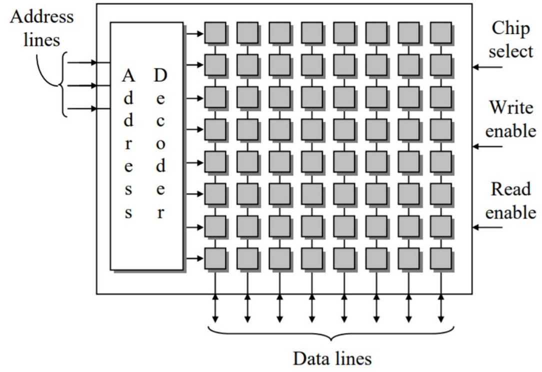
Basic Organization of a Memory Device

Memory Organization (Cont



- An **address decoder** selects exactly one row of the memory array to be active leaving the others inactive.
- The processor uses the inputs **read enable** and **write enable** to specify whether it is reading data from or writing data to the selected row of the memory array.
- When read enable is zero, we are reading data from memory, and when write enable is zero, we are writing data to memory. These two signals should never be zero at the same time.

Memory Organization (Cont



- If **latches** are used for the memory cells, then the **data lines** are connected to **the D inputs** of the **memory location latches** when data is written, and they are connected to the **Q outputs** when data is **read.**
- The **chip select** is an active low signal that enables and disables the memory device.
- If the chip select equals zero, the memory activates all of its input and output lines and uses them to transfer data.
- If the chip select equals one, the memory becomes idle, effectively disconnecting itself from all of its input and output lines

SRAM simulation

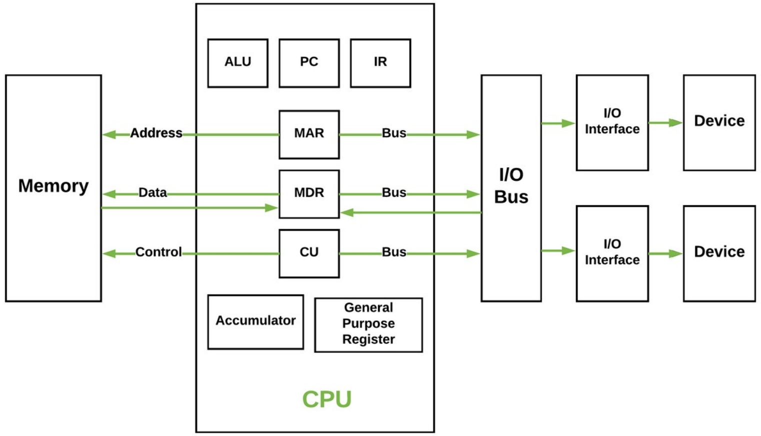
https://circuitverse.org/users/32582/projects/8b-addressable-ram

https://circuitverse.org/users/56463/projects/static-ram

Types of memory (DIY)

- SRAM
- DRAM
- ROM

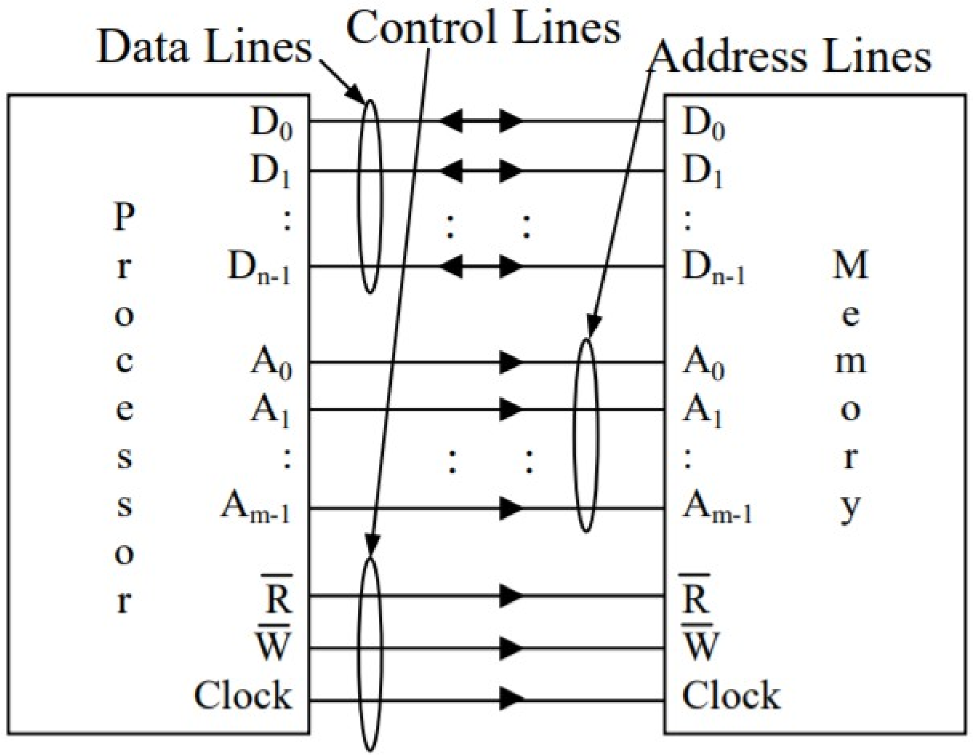
Interfacing Memory to a Processor



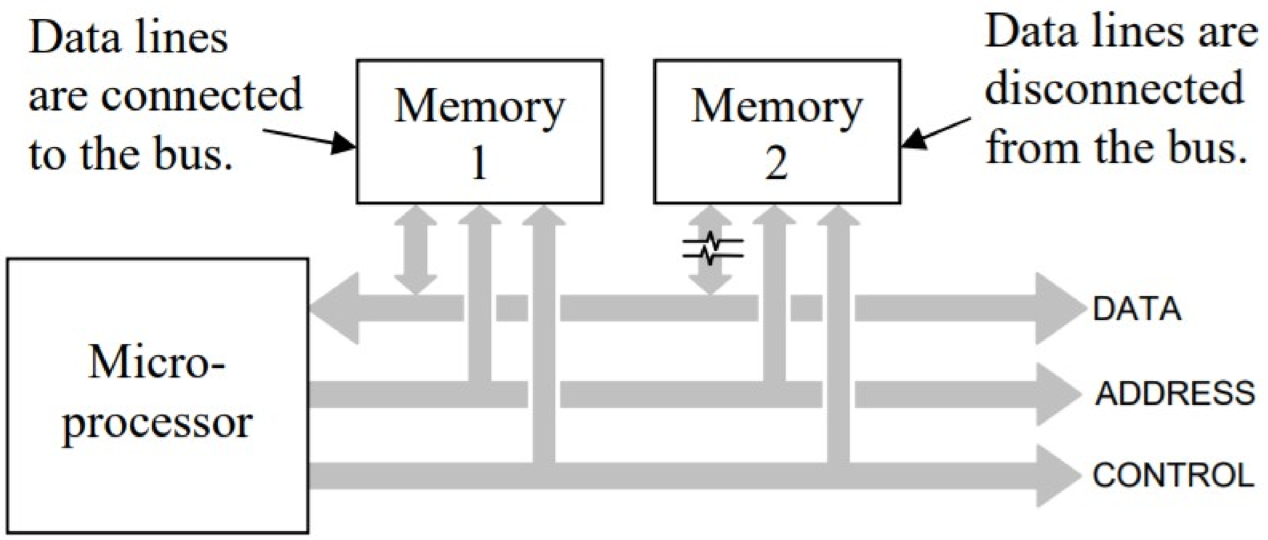
Buses

- In order to communicate with memory, a processor needs **three types of connections**: data, address, and control.
- **Data lines** are the electrical connections used to send data to or receive data from memory
- The **address lines** are controlled by the processor and are used to specify which memory location the processor wishes to communicate with.
- The **control lines** consist of the signals that manage the transfer of data.

Basic Processor to Memory Device Interface



Two Memory Devices Sharing a Bus

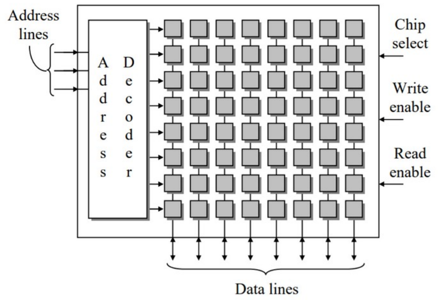


https://www.youtube.com/watch?v=QzWW-CBugZo

Memory Maps

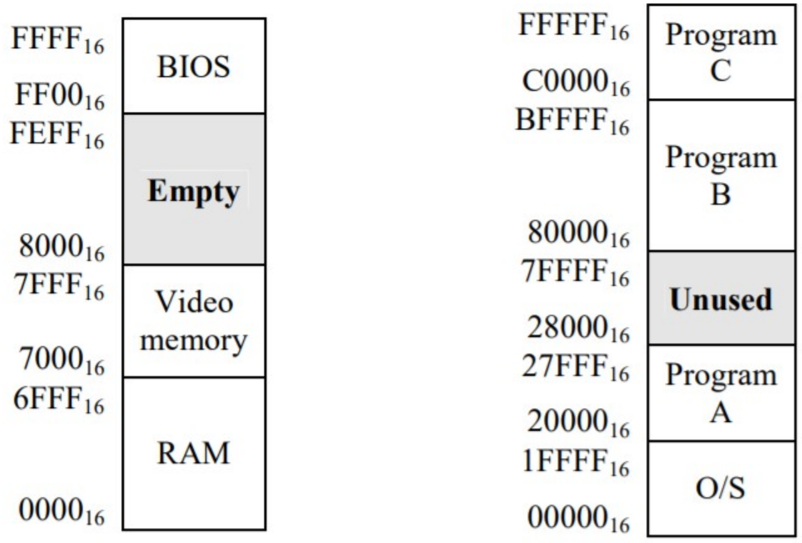
Memory Maps

- Think of memory as several filing cabinets where each folder can contain a single piece of data. The size of the stored data, i.e., the number of bits that can be stored in a single memory location, is fixed and is equal to the number of columns in the memory array
- Each piece of data can be either code (part of a program) or data (variables or constants used in the program). Code and data are typically stored in the same memory, each piece of which is stored in a unique address or row of memory



Memory map (Cont.)

- System designers describe the use of memory with a memory map. A memory map represents a system's memory with a long, vertical column. It is meant to model the memory array where the rows correspond to the memory locations.

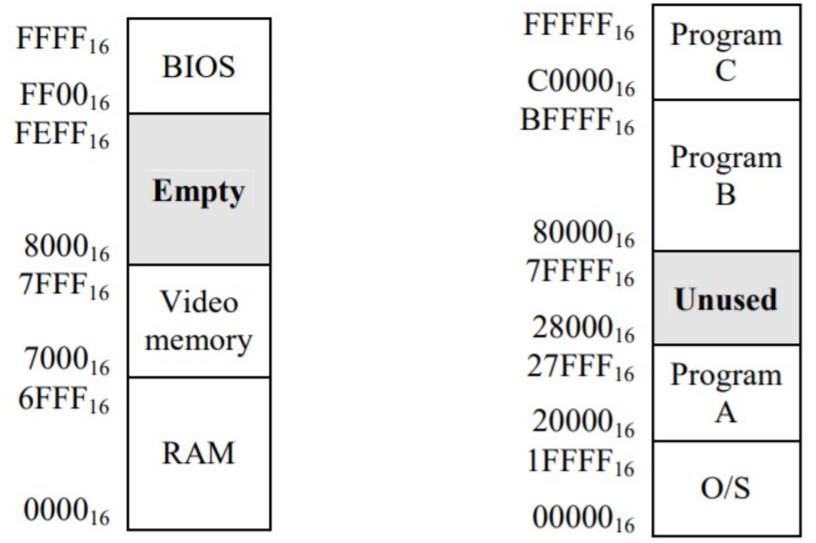


a.) Hardware-specific

b.) Software-specific

Memory space

- The memory map should represent the full address range of the processor. This full address range is referred to as the processor's **memory space**, and its size is represented by the number of memory locations in the full range, i.e., 2^m where m equals the number of address lines coming out of



a.) Hardware-specific b.) Software-specific

Address vs memory size

- Table presents a short list of memory sizes and the number of address lines required to access all of the locations within them.

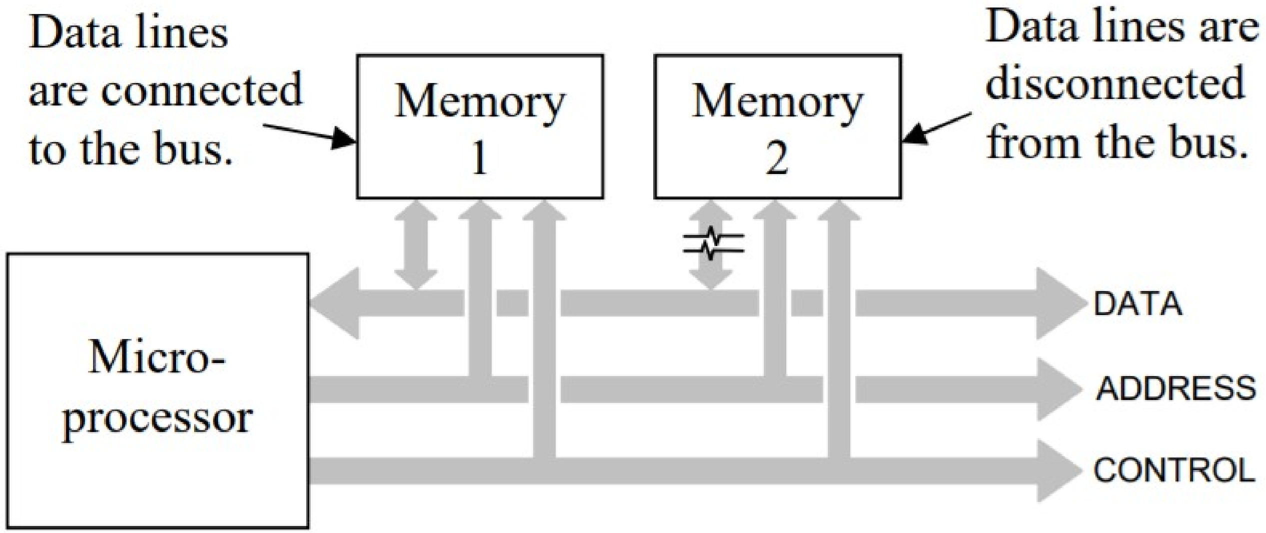
Table 12-2 Sample Memory Sizes versus Required Address Lines

Memory size	Number of address lines
1 K	10
256 K	18
1 Meg	20
16 Meg	24

Memory size	Number of address lines
256 Meg	28
1 Gig	30
4 Gig	32
64 Gig	36

Remember that the memory size is simply equal to 2^m where m is the number of address lines going into the device

Two Memory Devices Sharing a Bus



https://www.youtube.com/watch?v=QzWW-CBugZo

Address Decoding

- Address decoding is a method for using an address to enable a unique memory device while leaving all other devices idle.
- The bits of the full address are divided into two groups, one group that is used to identify the memory device and one group that identifies the memory location within the selected memory device.

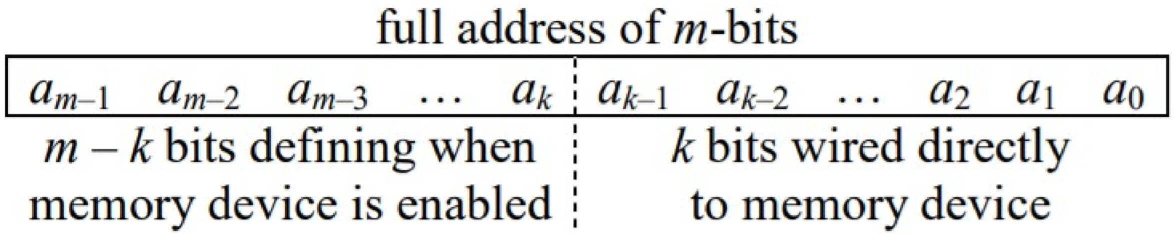


Figure 12-7 Full Address with Enable Bits and Device Address Bits

Example

a_{27}	a_{26}	a_{25}	a_{24}	a_{23}	a_{22}	…	a_2	a_1	a_0
4 bits that enable memory device				24 bits going to address lines of memory device					

- A processor with a 256 Meg address space is using the address 35E3C0316 to access a 16 Meg memory device.
  - How many address lines are used to define when the 16 Meg memory space is enabled?
  - What is the bit pattern of these enable bits that enables this particular 16 Meg memory device?
  - What is the address within the 16 Meg memory device that this address is going to transfer data to or from?
  - What is the lowest address in the memory map of the 16 Meg memory device?
  - What is the highest address in the memory map of the 16 Meg memory device?